

RS004 Week 9 — English Worksheet

Topic: Level Design (Data, Respawn, Finish) · **Course:** Platformer Game · **Time:** about 45 minutes

This week you think about how **level data** is kept apart from the game code. A `build_level()` function turns a list of numbers into platforms, the player **respawns** at the start if it falls off, and a **finish** marker says "you cleared it!". You will read this code closely and then explain the code you wrote in today's live lesson.

Keep these words handy: **level data**, `build_level()`, **respawn / start position**, **finish**, **cleared**.

1 · Predict 🧠

Read each snippet. Write what you think it does.

```
LEVEL_1 = [
    (0, 550, 800, 50),
    (150, 450, 120, 20),
]
def build_level(level_data):
    return [pygame.Rect(*p) for p in level_data]
```

What does `build_level(LEVEL_1)` give back? What does `*p` do?


```
if player.y > HEIGHT:
    player.x = start_x
    player.y = start_y
    velocity_y = 0
```

The player falls below the bottom of the screen. What happens to it?


```
if player.colliderect(finish):  
    cleared = True
```

Once **cleared** is **True**, what should change about the game?

2 · Why Separate the Data? 🤔

We keep the level as a list of numbers (`LEVEL_1`) instead of writing each `pygame.Rect(...)` by hand in the loop.

Give one reason this makes it easier to build *many* levels later.

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it on paper so it works, then explain why the original was wrong.

Bug A — `build_level` should return real platforms, but the game crashes saying it got tuples, not Rects.

```
def build_level(level_data):  
    return level_data
```

Hint: each tuple of numbers must be turned into a `pygame.Rect`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A player who falls off should respawn at the start. Right now it keeps falling forever and never comes back.

```
if player.y > HEIGHT:  
    velocity_y = 0
```

Hint: resetting velocity isn't enough — move the player back to its start spot.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Touching the finish should clear the level once. Right now the "Level Clear!" text never shows.

```
finish = pygame.Rect(710, 180, 50, 40)
if player.colliderect(player):
    cleared = True
```

Hint: the player is being compared with *itself*, not the finish.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Read this working level loader closely. It keeps the level as data, builds the platforms, respawns the player, and clears the level at the finish.

```
LEVEL_1 = [  
    (0, 550, 800, 50),  
    (150, 450, 120, 20),  
    (400, 350, 120, 20),  
]  
start_x, start_y = 50, 500  
  
def build_level(level_data):  
    return [pygame.Rect(*p) for p in level_data]  
  
platforms = build_level(LEVEL_1)  
finish = pygame.Rect(710, 180, 50, 40)  
cleared = False  
  
# inside the game loop:  
if player.y > HEIGHT:  
    player.x = start_x  
    player.y = start_y  
    velocity_y = 0  
  
if player.colliderect(finish):  
    cleared = True
```

What is stored inside `LEVEL_1`, and what does each group of four numbers describe?

What does `build_level(LEVEL_1)` return, and what is now inside `platforms`?

When does the `if player.y > HEIGHT:` block run, and what does it do to the player?

What has to be true for `cleared` to become `True` ?

Why is `cleared` set to `False` at the start instead of `True` ?

5 · Explain Your Lesson Code 🎥

Explain the code **you wrote in today's lesson**. Record a short video on your phone. You may show your level running. Try to use these words: **level data**, **build_level**, **respawn**, **finish**, **cleared**.

1. Show your level and point out the start and the finish.
2. Point to your level data and your `build_level()` and say what they do.
3. Fall off on purpose and show the respawn, then explain those lines.
4. Reach the finish and show the level clear.

Write what you will say in your video. Plan it here before you record.

Submit ✓

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.